

NGUYEN HOANG THANH PHAN

Waterloo, ON | 226-552-3205

phan7085@mylaurier.ca | linkedin.com/in/nguyen-hoang-thanh-phan | github.com/fanggodzzz

EDUCATION

Wilfrid Laurier University, Waterloo, Ontario

Bachelor of Science in Computer Science, Co-op Program

Relevant Coursework: Internet Computing, Computer Networks, Software Engineering, Database Systems, Operating Systems, Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence

September 2024 – August 2028

Cumulative GPA: 4.0/4.0

PROJECTS

AI-Powered Course Management Platform – Node.js, MySQL, Microsoft Azure, Google OAuth2.0

[GitHub](#) | July 2026

- Designed a scalable layered backend architecture to improve maintainability, enforce separation of concerns, and support future feature expansion.
- Implemented application security with JWT authentication and role-based authorization workflows, integrating Google OAuth APIs for secure Gmail-based user login
- Deployed and managed MySQL database infrastructure on Microsoft Azure, integrating database services with backend APIs for application data management
- Collaborated in a 12-week Agile Kanban team to manage sprint planning, develop features, conduct code reviews, and perform testing while ensuring timely delivery of software milestones.

To-Do List Web Application – Java, Spring Boot, MySQL, Docker, MS Azure

[GitHub](#) | [Demo](#) | April 2026

- Developed a full-stack task management web application using Spring Boot, delivering 10+ REST APIs with JWT authentication for secure user access.
- Implemented and deployed a MySQL database on Microsoft Azure, configuring secure environment variables and cloud database connectivity for production
- Developed 10+ unit and integration tests using JUnit and Mockito, improving application reliability by validating core business logic, API behavior, and end-to-end workflows.
- Containerized with Docker Compose, reducing multiple manual configuration to single-command deployment

Full-Stack-RAG: Software Documentation Retrieval System – Python, BM25, Word2Vec, Ollama

[GitHub](#) | August 2026

- Built an end-to-end RAG system for software documentation by developing automated crawling, preprocessing, retrieval, and response-generation pipelines in Python.
- Automated documentation processing by cleaning, normalizing, deduplicating, and chunking crawled content, producing structured datasets for downstream retrieval.
- Improved retrieval coverage by combining BM25 lexical matching with Word2Vec semantic similarity into a hybrid ranking pipeline for software development queries.
- Integrated Ollama into the retrieval pipeline to generate context-grounded responses from retrieved documentation, reducing reliance on unsupported model knowledge.

Dynamic Parking in Urban Networks – Python, NumPy, Reinforcement Learning

[GitHub](#) | April 2026

- Built a Q-learning-based parking allocation solution on weighted urban graph networks to optimize vehicle routing under dynamic traffic conditions
- Automated large-scale simulation experiments using Python, NumPy, and multiprocessing to execute parallel simulation runs, significantly improving data collection efficiency
- Analyzed experimental results by exporting simulation data to Excel and creating performance visualizations to evaluate learning behavior, convergence trends, and algorithm effectiveness

SKILLS

Languages: Java, Python, C/C++, SQL, HTML5, CSS3, JavaScript

Frameworks & Libraries: Spring Boot, Node.js, React

Databases: MySQL, Relational Database Design

Testing: JUnit, Mockito, Spring Boot Test, REST API Testing, Unit Testing, Integration Testing

AI Agents: Microsoft Copilot Pro, ChatGPT

Cloud & Deployment: Microsoft Azure, Render, Netlify, Vercel

DevOps & Tools: Maven, Docker, Docker Compose, Git, GitHub, Postman

AWARDS & ACHIEVEMENTS

Dean's Honour Roll - Faculty of Science, Wilfrid Laurier University

2024 – 2025

HackCanada Participant

March 2026

Developed a skin disease classification web app using FastAPI, React, and a machine learning model in 72 hours.